Department of Mathematics and Statistics
Course STAT2920: Introduction to probability
Quiz 2: Fall 2025

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Question 1. 5 pts

Let Y denote a random variable that takes on any of the values -2, 0, and 2 with respective probabilities

$$P(Y = -2) = 0.1, \quad P(Y = 0) = 0.6, \quad P(Y = 2) = 0.3.$$

Compute $\mathbb{E}\left[Y^2\right]$

Solution 1.

The second moment can be calculated as follow:

$$\begin{aligned}\mathbb{E}\left[Y^2\right] &= \sum_{y=\{-2,0,2\}} y^2 \cdot P(Y = y) \\ &= (-2)^2 P(Y = -2) + 0^2 \cdot P(Y = 0) + 2^2 \cdot P(Y = 2) \\ &= 0.4 + 1.2 = 1.6\end{aligned}$$

□

Question 2. 6 pts

If $\mathbb{E}[Z] = 1$ and $\mathbb{V}\mathrm{ar}(Z) = 3$, find

(a) (4 pts) $\mathbb{E}\left[(2 + Z)^2\right]$

(b) (2 pts) $\mathbb{V}\mathrm{ar}\,(4 + 3Z)$

Solution 2.

(a) First, let us calculate $\mathbb{E}\left[Z^2\right]$,

$$\mathbb{E}\left[Z^2\right] = \mathbb{V}\mathrm{ar}(Z) + \left[\mathbb{E}(Z)\right]^2 = 3 + 1 = 4. \qquad 2\,\text{pts}$$

Then,

$$\begin{aligned}\mathbb{E}\left[(2 + Z)^2\right] = \mathbb{E}\left[4 + 4Z + Z^2\right] &= 4 + 4\mathbb{E}\left[Z\right] + \mathbb{E}\left[Z^2\right] \\ &= 4 + 4 \times 1 + 4 = 12. \qquad 2\,\text{pts}\end{aligned}$$

(b) This can be done as

$$\mathbb{V}\mathrm{ar}\,(4 + 3Z) = 9\,\mathbb{V}\mathrm{ar}(Z) = 9 \times 3 = 27 \qquad 2\,\text{pts}$$

□

Question 3. 20 pts

Let X be a random variable with probability density function

$$f(x) = \begin{cases} c\left(1-|x|\right)^3, & -1 \le x \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

(a) (5 pts) What is the value of c?

(b) (5 pts) What is the cumulative distribution function of X?

(c) (5 pts) Calculate $\mathbb{E}[X]$

(d) (5 pts) Compute the variance $\mathbb{V}\mathrm{ar}(X)$

Solution 3.

(a) Since $f(x)$ is a valid pdf,

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

Thus,

$$1 = \int_{-1}^{1} c(1-|x|)^3\,dx.$$

Because the function is even,

$$1 = 2c\int_0^1 (1-x)^3\,dx = 2c\left[-\frac{(1-x)^4}{4}\right]_0^1 = 2c\left(\frac{1}{4}\right) = \frac{c}{2}.$$

Therefore,

$$\boxed{c = 2.}$$

(b) For $x < -1$:

$$F(x) = 0.$$

For $-1 \le x < 0$:

$$F(x) = \int_{-1}^{x} 2(1-|t|)^3\,dt = \int_{-1}^{x} 2(1+t)^3\,dt.$$

Integrating,

$$F(x) = 2\left[\frac{(1+t)^4}{4}\right]_{-1}^{x} = \frac{(1+x)^4}{2}.$$

For $0 \le x < 1$:

$$F(x) = F(0) + \int_0^x 2(1-t)^3\,dt.$$

Since $F(0) = \frac{(1+0)^4}{2} = \frac{1}{2}$,

$$F(x) = \frac{1}{2} + 2\left[-\frac{(1-t)^4}{4}\right]_0^x = \frac{1}{2} + \frac{1}{2} - \frac{(1-x)^4}{2} = 1 - \frac{(1-x)^4}{2}.$$

For $x \ge 1$:

$$F(x) = 1.$$

Hence,

$$F(x) = \begin{cases} 0, & x < -1, \\ \dfrac{(1+x)^4}{2}, & -1 \le x < 0, \\ 1 - \dfrac{(1-x)^4}{2}, & 0 \le x < 1, \\ 1, & x > 1. \end{cases}$$

(c) Since $f(x)$ is even and $xf(x)$ is odd,

$$\mathbb{E}[X] = \int_{-1}^{1} x f(x)\, dx = 0.$$

Thus,

$$\boxed{\mathbb{E}[X] = 0.}$$

(d) We have

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathbb{E}[X^2].$$

Compute

$$\mathbb{E}[X^2] = \int_{-1}^{1} x^2 f(x)\, dx = 2\int_{0}^{1} x^2(2)(1-x)^3\, dx = 4\int_{0}^{1} x^2(1-x)^3\, dx.$$

Expand and integrate:

$$\int_0^1 x^2(1-x)^3 dx = \int_0^1 (x^2 - 3x^3 + 3x^4 - x^5)\, dx = \frac{1}{3} - \frac{3}{4} + \frac{3}{5} - \frac{1}{6} = \frac{1}{60}.$$

Hence,

$$\mathbb{E}[X^2] = 4 \times \frac{1}{60} = \frac{1}{15}.$$

Thus,

$$\boxed{\mathrm{Var}(X) = \frac{1}{15}.}$$

□

Question 4. 10 pts

The probability mass function of a random variable X is given by

$$f(x) = \frac{k \cdot 2^x}{x!}, \quad x = 0, 1, 2, \ldots$$

(a) (4 pts) Find the mgf of X

(b) (6 pts) Use the mgf to calculate $\mathbb{E}[X]$ and $\mathbb{V}\mathrm{ar}(X)$

Solution 4.

(a) First we need to find k

$$\sum_{x=0}^{\infty} f(x) = 1 \Leftrightarrow \sum_{x=0}^{\infty} \frac{k \cdot 2^x}{x!} = 1$$
$$\Leftrightarrow \quad k \underbrace{\sum_{x=0}^{\infty} \frac{2^x}{x!}}_{e^2} = 1$$
$$\Leftrightarrow \quad k \cdot e^2 = 1$$
$$\Leftrightarrow \quad k = \frac{1}{e^2} = e^{-2}$$

Thus, $f(x) = \frac{e^{-2} 2^x}{x!}$ (1.5 pts).

Therefore, the mgf is given by

$$M_X(t) = \mathbb{E}\left[e^{tX}\right] = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-2} 2^x}{x!}$$
$$= e^{-2} \underbrace{\sum_{x=0}^{\infty} \frac{\left(2e^t\right)^x}{x!}}_{=e^{2e^t}}$$
$$= e^{2\left(e^t - 1\right)}, \qquad t \in \mathbb{R}. \qquad 2.5 \text{ pts}$$

(b) Taking derivatives,

$$M_X'(t) = 2e^t e^{2\left(e^t-1\right)} \qquad 1 \text{ pts}$$
$$M_X''(t) = 2e^t e^{2\left(e^t-1\right)} \left(2e^t + 1\right) \qquad 1.5 \text{ pts}$$

Therefore,

$$\mathbb{E}[X] = M_X'(0) = 2e^0 e^{2\left(e^0-1\right)} = 2 \qquad 1 \text{ pts}$$
$$\mathbb{E}\left[X^2\right] = M_X''(0) = 2e^0 e^{2\left(e^0-1\right)} \left(2e^0 + 1\right) = 2\left(2e^0 + 1\right) = 6 \qquad 1 \text{ pts}$$

and hence

$$\mathbb{V}\text{ar}(X) = \mathbb{E}\left[X^2\right] - \left(\mathbb{E}[X]\right)^2 = 6 - 2^2 = 2. \qquad 1.5 \text{ pts}$$

□